Optimization Algorithm Design via Electric Circuits

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Optimization problem formulation

$$\begin{array}{ll} \text{minimize} & f(x) \\ \text{subject to} & x \in \mathcal{R}(E^\intercal), \end{array}$$

where $x \in \mathbb{R}^m$ is the optimization variable, $f : \mathbb{R}^m \to \mathbb{R} \cup \{\infty\}$ is closed, convex, and proper, and $E \in \mathbb{R}^{n\times m}$. Assume we have n nets $N_1, \ldots, N_n$ forming a partition of $\{1, \ldots, m\}$. More specifically, we let $E \in \mathbb{R}^{n\times m}$ be a *selection matrix* defined as

$$E_{ij} = \begin{cases} +1 & \text{if } j \in N_i \\ 0 & \text{otherwise.} \end{cases}$$

Our goal is to find a primal-dual solution satisfying the KKT conditions

$$y \in \partial f(x), \quad x \in \mathcal{R}(E^\intercal), \quad y \in \mathcal{N}(E).$$

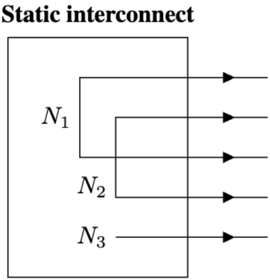
Example

$$\begin{array}{ll} \underset{x_1,\ldots,x_N \in \mathbf{R}^{m/N}}{\text{minimize}} & f_1(x_1) + \cdots + f_N(x_N) \\ \text{subject to} & x_1 = \cdots = x_N, \end{array}$$

where $x = (x_1, \ldots, x_N)$ is the decision variable, the objective function $f(x) = f_1(x_1) + \cdots + f_N(x_N)$ is block-separable, and $E^T = (I, \ldots, I) \in \mathbb{R}^{m \times m/N}$.

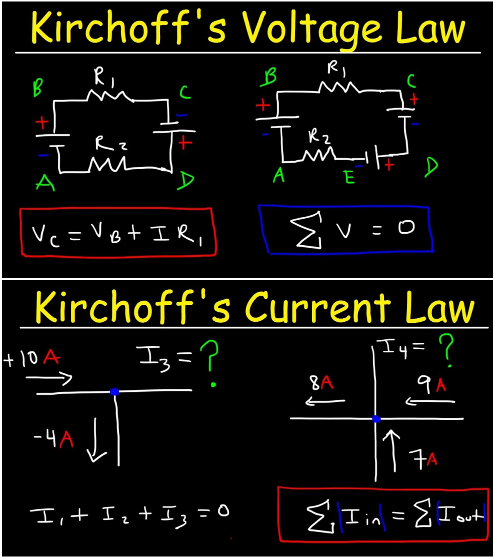
Static interconnect

The static interconnect is a set of (ideal) wires connecting m terminals and forming n nets. Let $x \in \mathbb{R}^m$ be a vector of terminal potentials and $y \in \mathbb{R}^m$ be a vector of currents leaving the terminals. Using matrix $E \in \mathbb{R}^{n \times m}$, we can express Kirchhoff's voltage law (KVL) as $x \in \mathcal{R}(E^\mathsf{T})$ and Kirchhoff's current law (KCL) as $y \in \mathcal{N}(E)$. In other words, the static interconnect enforces the V-I relationship



$$(x, y) \in \mathcal{R}(E^\mathsf{T}) \times \mathcal{N}(E).$$

Especially for ML engineers

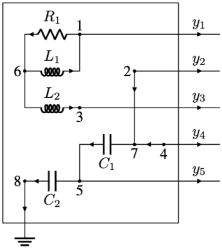


Dynamic interconnect

Denote the number of nodes in the RLC circuit by τ. Connect nodes $1, 2, \ldots, m$ to terminals $1, 2, \ldots, m$, and let the last node, node τ, be the ground node. (This implies $\tau \geq m + 1$.) Denote the number of RLC components by σ. We describe the topology with a reduced node incidence matrix (with the bottom row corresponding to the ground node removed) $A \in \mathbb{R}^{(\tau-1)\times\sigma}$ defined as

$$A_{ij} = \begin{cases} +1 & \text{if node } i \text{ connects to } + \text{ terminal of component } j \\ -1 & \text{if node } i \text{ connects to } - \text{ terminal of component } j \\ 0 & \text{otherwise.} \end{cases}$$

Dynamic interconnect



$$A = \begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \\ 7 \end{array} \begin{array}{c} \begin{array}{ccccccc} R_1 & R_2 & R_3 & L_1 & L_2 & C_1 & C_2 \end{array} \\ \begin{bmatrix} +1 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & +1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & +1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & +1 \\ -1 & 0 & 0 & +1 & +1 & 0 & 0 \\ 0 & -1 & -1 & 0 & 0 & +1 & 0 \end{bmatrix} \end{array}$$

Dynamic interconnect

Let $x \in \mathbb{R}^m$ be the potentials at the m terminals, which are connected to nodes $1, \ldots, m$, and $y \in \mathbb{R}^m$ be the currents leaving the terminals. Denote the node potential vector with the ground node excluded (since the potential at ground is 0) by

$$\begin{bmatrix} x \\ e \end{bmatrix} \in \mathbb{R}^{\tau-1}.$$

$$\text{(i) } Ai = \begin{bmatrix} -y \\ 0 \end{bmatrix} \quad \text{(KCL)} \qquad \text{(ii) } v = A^\intercal \begin{bmatrix} x \\ e \end{bmatrix} \quad \text{(KVL)}$$

$$\text{(iii) } v_{\mathcal{R}} = D_{\mathcal{R}} i_{\mathcal{R}} \quad \text{(Resistor)} \qquad \text{(iv) } v_{\mathcal{L}} = D_{\mathcal{L}} \frac{d}{dt} i_{\mathcal{L}} \quad \text{(Inductor)}$$

$$\text{(v) } i_{\mathcal{C}} = D_{\mathcal{C}} \frac{d}{dt} v_{\mathcal{C}} \quad \text{(Capacitor)}$$

where $D_{\mathcal{R}}$, $D_{\mathcal{L}}$, and $D_{\mathcal{C}}$ are diagonal matrices respectively with resistances, inductances, and capacitances values in the diagonals.

Admissibility

When an RLC circuit reaches equilibrium, voltages across inductors and currents through capacitors are 0. We say a dynamic interconnect is *admissible* if it relaxes to the static interconnect at equilibrium. Mathematically, this condition is expressed as

$$\left\{(x,y)\,\middle|\, Ai = \begin{bmatrix} -y \\ 0 \end{bmatrix}, v = A^\intercal \begin{bmatrix} x \\ e \end{bmatrix}, v_{\mathcal{R}} = D_{\mathcal{R}} i_{\mathcal{R}}, v_{\mathcal{L}} = 0, i_{\mathcal{C}} = 0\right\}$$